

Exercise Sheet 3

Fundamentals

Automatic Differentiation

Exercise 3.1: Computational Graphs

optional

Draw the computational graph for the function:

$$f(x, y, z) = \frac{(x \cdot y) \sqrt{z}}{\exp(x)} \quad (1)$$

Write down some of the derivatives that you could compute with this graph (you do not need to compute the derivatives, just discuss which ones could be calculated and why).

Exercise 3.2: Backpropagation

optional

Explain the purpose of the backpropagation algorithm.

Exercise 3.3: Gradient Descent

optional

Explain the idea behind the gradient descent algorithm. Write down the update formula. What would happen if the sign was inverted?

Practical: Training the CARLA Baseline Models

In this exercise you train the three models that will be the subject of your safety case. Use the provided CARLA dataset.

The dataset contains forward-facing camera images collected in CARLA under *clear weather* and *daytime* conditions, each annotated with three binary labels: *traffic light present*, *pedestrian present*, *vehicle present*.

You will train **three separate binary classifiers** – one for each detection task:

- A **pedestrian detector**
- A **traffic light detector**
- A **vehicle detector**

Each model takes a camera image as input and outputs a single binary prediction, trained with cross-entropy (CE) loss.

Exercise 3.4: Dataset Exploration

Load the dataset and explore it.

1. How many images are in the training and test splits?
2. What is the class distribution for each label? Are the classes balanced?
3. Display a few example images for each label combination. What patterns do you notice?

Exercise 3.5: Train Three Binary Classifiers

Train three separate binary classifiers on the training split – one for each detection task (pedestrian, traffic light, vehicle). Use Python and preferably **PyTorch**. You may use a pre-trained backbone (e.g., ResNet-18) and fine-tune it for binary classification. Commit your code to your **git** repository.

1. Describe your model architecture and training setup (optimizer, learning rate, loss function).
2. Train each model and plot the training and validation loss curves. Do the models converge?
3. Why might training three separate models be preferable to a single multi-label classifier from a safety perspective?

Exercise 3.6: Evaluation

Evaluate each of your three trained models on the test split.

1. Report accuracy, precision, recall, and F1-score for each model.
2. Which model performs worst? Hypothesize why.
3. From a safety perspective, which metric matters most for each model – precision or recall? Explain.

Save all three trained models. You will use them in all subsequent exercise sheets.

ODD Gap Analysis

Exercise 3.7: Comparing the ODD to the Training Data

In Exercise Sheet 2 you defined an Operational Design Domain (ODD) based on the system description – before seeing any data. Now that you have explored the dataset and trained your models, revisit that ODD.

1. Look at the data you trained on. What conditions does it cover? List the environmental conditions represented in the training set (weather, lighting, scene types, etc.).
2. Compare this to the ODD you defined in Exercise Sheet 2. Which ODD dimensions are *not* covered by the training data?
3. What are the implications of this gap for your models' safety? For which ODD conditions can you *not* make any claims about model performance?
4. Note down the gaps you identified (if any). You will revisit them in Sheets 4 and 5 when you formally measure ODD coverage and refine the ODD.